

Irrationality of Cubic Threefolds after One Stabilization

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Abstract

We prove that $X \times \mathbf{P}^1$ is irrational for every smooth complex cubic threefold X . Our invariant counts certain rank-two blocks of the generic even quantum D -module. A counted block has centered leading term $N \neq 0$ with $N^2 = 0$, and after a canonical elementary modification its two residue eigenvalues are distinct modulo $\mathbf{Z}$. Iritani's blowup decomposition and weak factorization make this count birationally invariant because every point, curve, and surface center contributes zero to the blowup formula. Iritani–Koto's projective-bundle decomposition computes its value after stabilization; the resulting count distinguishes $X \times \mathbf{P}^1$ from $\mathbf{P}^4$. The same count gives an irrationality criterion for smooth projective threefolds, the one-stabilization theorem for prime Fano threefolds of genus eight, and irrational universally CH_0 -trivial fourfolds.

Keywords. cubic threefold; universal CH_0 -triviality; irrationality; quantum D -modules; formal exponents; birational invariants; weak factorization; Fano threefolds.

1 Introduction

Let X be a smooth complex cubic threefold. Clemens and Griffiths proved that X is irrational [8], but their intermediate-Jacobian argument does not directly obstruct the rationality of $X \times \mathbf{P}^1$: in a weak factorization of a birational map between smooth fourfolds, the blowup formula for a surface center S contributes $H^3(S)(-1)$ in degree five, while $H^3(X)(-1)$ occurs in $H^5(X \times \mathbf{P}^1)$. We prove that the stabilized fourfold is nevertheless irrational.

Theorem 1.1 (One-step irrationality). *For every smooth complex cubic threefold X , the fourfold $X \times \mathbf{P}^1$ is irrational.*

The stable-rationality picture varies across the moduli space. Hassett and Tschinkel identified varieties birational to cubic threefolds as the exceptional Fano-threefold case not covered by their stable-irrationality theorem [11]. More recently, Engel, de Gaay Fortman, and Schreieder proved that a very general complex cubic threefold is not stably rational [10, Corollary 1.4], whereas Tschinkel and Zhang constructed explicit stably rational smooth cubic threefolds over $\mathbf{Q}$ [21, Theorem 1.3]. The theorem above applies uniformly in both situations: it also proves that the first stabilization of those stably rational examples is irrational. A companion paper proves that this bound is sharp: it gives two explicit examples for which $X \times \mathbf{P}^2$ is rational [20].

The proof uses the quantum D -module (QDM): the cohomology module over the ring of Novikov and bulk parameters, equipped with its flat quantum connection [16]. The Novikov parameters record curve classes, while the bulk parameters are coordinates on cohomology. By the even QDM we mean that the odd bulk variables are set to zero and the module is restricted to even cohomology. We then replace the ring of numerical Novikov and even

bulk parameters by its fraction field; this is what *generic* means below. In the direction of the connection variable z , the leading coefficient is quantum multiplication by the Euler field. Over an algebraic closure of the generic coefficient field, its generalized eigenspaces decompose the formal connection into blocks.

We record the block isomorphism classes in a free commutative monoid. For an additive marker on this monoid, Iritani’s blowup theorem gives a formula for its change under blowup, with each center summand retained. Combined with weak factorization, this formula makes any such marker that vanishes on every center occurring in a weak factorization of smooth fourfolds a birational invariant.

The marker counts rank-two blocks for which the centered leading term N of Euler multiplication is nonzero and satisfies $N^2 = 0$. Horizontality of the Poincaré pairing makes a canonical elementary modification regular singular, while flatness gives a Lax equation for its residue, whose eigenvalues are therefore preserved. We count the block when the two residue eigenvalues are distinct modulo $\mathbf{Z}$, equivalently when the two formal exponent classes are distinct. Using Beauville’s quantum product formulas, we compute the exponent representatives $-1/6$ and $-5/6$ for the cubic block, with difference $2/3$. The resulting marker vanishes on points, curves, and surfaces and on projective space. Iritani–Koto’s projective-bundle theorem shows that it doubles after product with $\mathbf{P}^1$. This proves the theorem.

This use of Euler eigenspace blocks in birational geometry belongs to the same circle of ideas as the Hodge-atom framework of Katzarkov–Kontsevich–Pantev–Yu [17]. Their framework combines the generalized eigenspaces of Euler multiplication, Iritani’s blowup theorem, and weak factorization to produce birational invariants. In particular, their Example 6.21 uses the zero spectral piece, enhanced by the Serre operator and formal monodromy, to recover the irrationality of every smooth cubic threefold [17, Example 6.21]. Recent applications and refinements include work on cubic, Gushel–Mukai, and Küchle fourfolds [12, 6, 5].

The marker used here retains less structure: it uses no Hodge structure and records only the rank-two formal-exponent condition above. Its purpose is the one-stabilization problem. The additional ingredients are a specialization that keeps the center summands distinct in Iritani’s blowup comparison, the vanishing of the marker on points, curves, and surfaces, and Iritani–Koto’s projective-bundle comparison for stabilization [14, 15]. Gyenge gives a complementary formal tensor-product theorem for quantum D -modules of products and proves multiplicativity for Hodge atoms [13]. Cai independently computes the same two exponent representatives and uses them with Iritani’s blowup theorem to prove symplectic irrationality of cubic threefolds [7]; that argument does not address stabilization.

The same construction gives a threefold criterion without stabilization.

Corollary 1.2 (Threefold marker criterion). *Let Y be a smooth projective complex threefold. If its generic even QDM contains a block counted by the formal-exponent marker defined in (2.12), then Y is irrational.*

Indeed, the only centers in a weak factorization of threefolds have dimension at most one, where the marker vanishes, while the generic quantum product of $\mathbf{P}^3$ is semisimple. This gives an irrationality criterion formulated in terms of the quantum D -module. We do not claim that this marker coincides with a semiorthogonal or derived-categorical rationality obstruction.

A further application comes from Kuznetsov’s construction. For an arbitrary smooth prime Fano threefold V of genus eight, it associates a smooth Pfaffian cubic threefold X and rank-two projective bundles over V and X related by a flop [18, Theorems 2.17–2.18].

Tschinkel and Zhang study equivariant and twisted stable-birational refinements of this Pfaffian–Grassmannian correspondence [22]. This gives the following consequence.

Corollary 1.3 (Irrationality after one stabilization for V_{14}). *For every smooth prime Fano threefold V of genus eight, $V \times \mathbf{P}^1$ is irrational.*

The theorem also yields universally CH_0 -trivial irrational fourfolds. Universal CH_0 -triviality passes to a projective bundle by the projective-bundle formula for Chow groups.

Corollary 1.4 (Separation in codimension at most three). *There is a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds such that, for every member X , both X and $X \times \mathbf{P}^1$ are universally CH_0 -trivial, while $X \times \mathbf{P}^1$ is irrational.*

Corollary 1.5 (Fermat cubic). *Let*

$$X = \{x_1^3 + x_2^3 + x_3^3 + x_4^3 + x_5^3 = 0\} \subset \mathbf{P}^4.$$

Then the fourfold $X \times \mathbf{P}^1$ is universally CH_0 -trivial and irrational.

Corollary 1.6 (Coprime-degree cubics). *Let X be one of the cubic threefolds of [24, Theorem 3.3]. Then $X \times \mathbf{P}^1$ admits unirational parametrizations of coprime degrees, is universally CH_0 -trivial, and is irrational.*

Voisin proves the codimension-at-most-three statement [23, Theorem 4.5]. The Fermat case follows from Colliot-Thélène’s separated-variable criterion [9]. Yang–Yu–Zhu construct the coprime-degree family [24, Theorems 1.1 and 3.3]. The last corollary answers the $m = 1$ case of their question; it does not address all stabilizations.

Section 2 constructs the block monoid, proves the marker formulas for blowups and projective bundles, defines the exponent marker, computes the cubic block, and proves that the marker vanishes on centers of dimension at most two. Section 3 gives the threefold, genus-eight, and universal- CH_0 consequences.

2 QDM markers and one stabilization

We prove the theorem by applying the quantum D -module decomposition formulas directly. The common argument is stated first for an arbitrary additive marker. We then construct the specific marker used for a cubic threefold and prove that it vanishes on points, curves, and surfaces.

Here is the concrete invariant before the general construction. Split the generic even quantum D -module into formal connection blocks. A block contributes one if it has rank two, its centered leading term N is nonzero with $N^2 = 0$, and the two residue eigenvalues after the canonical elementary modification are distinct modulo $\mathbf{Z}$. These eigenvalues represent the block’s two formal exponent classes. Every other block contributes zero, and the sum of the contributions is I_{exp} . The rest of the section shows that this count is intrinsic and additive, vanishes on points, curves, and surfaces, equals one on a cubic threefold, doubles after product with $\mathbf{P}^1$, and vanishes on $\mathbf{P}^4$.

2.1 Block multisets and a single blowup

Let Y be smooth and projective. We first make three setup choices. Pass to numerical Novikov variables, set the odd bulk variables to zero, and restrict the underlying cohomology space to $H^{\text{ev}}(Y)$. The last restriction is essential: restricting the bulk base alone would

still leave odd cohomology in the fibre. Let K_Y be the generic coefficient field of this even QDM and fix an algebraic closure $\overline{K}_Y$. We form all spectral blocks only after base change to $\overline{K}_Y$, so no later scalar extension can split a factor that has already been counted as one block.

Let U denote multiplication by the Euler field on $H^{\text{ev}}(Y)$. After base change to $\overline{K}_Y$, each generalized eigenspace of U is a summand of the full formal connection, not just an invariant subspace of U :

Proposition 2.1 (Formal spectral connection splitting). *Suppose the leading operator of a finite free formal QDM has block-diagonal form $U_i = \lambda_i I + N_i$, where each N_i is nilpotent and $\lambda_i - \lambda_j$ is a unit for $i \neq j$. There is a unique gauge $G = I + O(z)$, regular in z and block off-diagonal in positive orders, which makes the z -connection block diagonal. In this gauge every bulk and Novikov direction is block diagonal as well. The horizontal pairing is block diagonal and nondegenerate on each factor.*

Proof. At order z^{n-1} , the (i, j) -entry of the gauge equation is a Sylvester equation

$$U_i X - X U_j = B_{n,ij}.$$

Its operator is $(\lambda_i - \lambda_j)I$ plus the nilpotent operator $X \mapsto N_i X - X N_j$, hence is invertible. Recursion gives the normalized gauge, and the same equation with zero right-hand side proves uniqueness.

Choose frames of the generalized-eigenspace subbundles after the splitting extension. If an off-diagonal block of a base-direction connection had a first nonzero Laurent coefficient Xz^m , the lowest off-diagonal coefficient of flatness would be $U_i X - X U_j = 0$, contrary to Sylvester invertibility. Thus every base direction preserves the same factors. Applied coefficientwise to pairing horizontality, the identical argument kills the off-diagonal pairing blocks. Nondegeneracy of the total pairing then makes every diagonal restriction nondegenerate. $\square$

We call these factors the *generic even QDM blocks* of Y . A block retains its connection, pairing, and centered Euler endomorphism, but not an absolute cohomological grading. A block type may retain additional marked data. If it retains grading, a morphism may carry a recorded integral suspension, and the marker must be invariant under that suspension. The block type used below forgets absolute grading. Its morphisms preserve the retained data and are regular connection isomorphisms after scalar extension and formal change of bulk coordinates. In particular, their gauges use neither negative nor fractional powers of the connection variable z .

Fix a block type $\mathcal{T}$. Let $\Pi_{\mathcal{T}}$ be the regular-isomorphism groupoid of blocks of that type and put

$$\mathcal{L}_{\mathcal{T}} := \pi_0 \text{Sym}^{\sqcup}(\Pi_{\mathcal{T}}),$$

the free commutative monoid on their isomorphism classes, equivalently the monoid of finite multisets of such classes. Direct sum gives each Y its block multiset $\mathcal{B}_{\mathcal{T}}(Y) \in \mathcal{L}_{\mathcal{T}}$. An *additive marker* is a commutative-monoid homomorphism

$$w : \mathcal{L}_{\mathcal{T}} \longrightarrow A, \quad I_w(Y) := w(\mathcal{B}_{\mathcal{T}}(Y)),$$

where A is a commutative monoid. We do not pass to a group completion: subtraction would destroy useful markers, including Boolean ones.

We use the following notation.

Consider one blowup $\beta : \tilde{Y} = \text{Bl}_Z Y \rightarrow Y$ along a smooth center of codimension $c \geq 2$. Iritani's comparison has one ambient summand and $c - 1$ center summands. Distinct copies

Symbol	Meaning
$\Pi_{\mathcal{T}}$	regular-isomorphism groupoid of typed QDM blocks
$\mathcal{L}_{\mathcal{T}}$	$\pi_0 \text{Sym}^{\sqcup}(\Pi_{\mathcal{T}})$, finite block multisets
$\mathcal{B}_{\mathcal{T}}(Y)$	multiset of block isomorphism classes for Y
$w : \mathcal{L}_{\mathcal{T}} \rightarrow A$	additive marker with values in A
ω_j	data attached to the j -th center summand
$c_w(\omega_j)$	marker contribution of that center summand

Table 1: Notation for the block monoid and center summands.

of the same center must remain distinct in the blowup formula. We therefore retain separate data for each center summand: for the j -th, record

$$\omega_j = (\beta, Z, j, \varsigma_j, \chi_j), \quad 0 \leq j \leq c-2,$$

where ς_j is Iritani's center reconstruction parameter and χ_j is the induced numerical Novikov specialization. Its correction is $c_w(\omega_j) := w(\mathcal{B}_{\mathcal{T}}(Z; \omega_j))$. The comparison will give equality only after applying w ; distinct block multisets may have the same marker.

Theorem 2.2 (Birational invariance of additive markers). *Let $w : \mathcal{L}_{\mathcal{T}} \rightarrow A$ be an additive marker unchanged by the allowed regular scalar extensions and formal coordinate changes. Suppose*

$$I_w(\mathbf{P}_Y(V)) = rI_w(Y), \quad \text{rk } V = r, \quad (2.1)$$

$$I_w(\text{Bl}_Z Y) = I_w(Y) + \sum_{j=0}^{c-2} c_w(\omega_j), \quad \text{codim}_Y Z = c \geq 2. \quad (2.2)$$

If every summand associated with a smooth center of dimension at most $d-2$ has zero correction, then I_w is a birational invariant of smooth projective d -folds.

Proof. Projective weak factorization writes a birational map of smooth projective d -folds as a zigzag of blowups and blowdowns along smooth centers of codimension at least two [1, Theorem 0.3.1]. By (2.2) and the vanishing of every center correction, the marker is unchanged at each step; the equalities therefore telescope. Disconnected centers are treated componentwise, while a divisor blowup is an isomorphism. $\square$

Equivalently, the marker descends on object sets:

$$\begin{array}{ccc} \text{Ob}(\text{SmProj}_d) & \xrightarrow{I_w} & A \\ \downarrow & & \parallel \\ \text{Ob}(\text{SmProj}_d)/\sim_{\text{bir}} & \xrightarrow{\bar{I}_w} & A. \end{array} \quad (2.3)$$

This square reformulates the proof; it is not an additional input.

2.2 Compatibility with blowups and projective bundles

To use the cited decomposition theorems, we must check that they preserve all the data seen by the marker. Four issues arise: coefficient fields, restriction to even cohomology, regularity in z , and faithful embedding of the center coefficients. The last issue is the delicate one: Iritani's center Novikov map may have a kernel, so the independent target coordinates must recover the missing numerical curve-class data.

Coefficient fields. The intrinsic ample-adic completion and the Laurent neighborhood of a fibre or exceptional cusp need not map to one another. Iritani’s display (1.1) supplies the required common coefficient maps [14, (1.1)]; injectivity for each center summand is established below. For projective bundles the common source is $\mathbf{C}[z, q][[Q, \hat{\tau}]]$. On the blowup side the ambient term uses the corresponding graded source. A center term begins with its intrinsic reduced coefficient ring from Remark 2.3, maps to the possibly smaller image ring of Remark 5.6, and is then pulled to a separate target ring for that summand in the external direct sum. Embed the generic fields of the ambient common image and of these separate center images in a fixed algebraically closed overfield and form every primary block there. Base change after this splitting extension preserves the rank, nilpotent Jordan type, regular formal gauge, and formal exponent classes of each block; forming primary factors over a nonsplitting field before extension would not.

Restriction to even cohomology. The comparison is restricted to even cohomology, not merely to the even bulk base. The supermodule conventions and the pull-push and Fourier formulas in Iritani’s Section 5.8 and Iritani–Koto’s construction are parity-preserving. After the odd bulk variables are set to zero, the comparison and its inverse therefore restrict to mutually inverse maps on H^{ev} . This is the QDM on which the additive marker is defined.

Regularity in z . The comparisons are regular in z . Iritani’s Theorem 5.18 and Iritani–Koto’s Theorem 5.1 are respectively defined over rings of the form

$$\mathbf{C}[z]((q_{\text{exc}}^{-1/\mathfrak{s}}))[[Q, \hat{\tau}]], \quad \mathbf{C}[z]((q_{\text{fib}}^{-1/r'}))[[Q, \hat{\tau}]],$$

where Iritani–Koto take $r' = r$ when $r - 1$ is even and $r' = 2r$ when $r - 1$ is odd. Here $\mathfrak{s} \in \{c - 1, 2(c - 1)\}$ is Iritani’s parity-dependent ramification index from [14, (5.11)]. Their inverses use no negative powers of z ; see also [15, Remark 5.3]. The maps are homogeneous but need not have degree zero: the projective-bundle map has degree $-(r - 1)$, while Iritani’s ambient and center components have degrees 0 and $-c$, respectively [15, Theorem 5.1(3)][14, Theorem 5.18(3)]. Because the block type used here does not retain absolute grading, these shifts do not change its component. For a graded refinement they are recorded as integral suspensions, and the operation formula applies only to a marker invariant under those suspensions.

Faithful center coordinates. Iritani’s center map need not be injective [14, (5.15)]. After numerical reduction it has the form

$$Q_Z^d \mapsto Q^{i_* d} q_{\text{exc}}^{-\rho_Z \cdot d / (c - 1)}. \quad (2.4)$$

Here $\rho_Z = c_1(N_{Z/Y})$. The raw map is not the coefficient map used below. Remark 2.3 gives the intrinsic divisor-equation-reduced coefficient ring, whose Novikov generators are the combined symbols

$$X_d = Q_Z^d \exp(\sigma^{(2)} \cdot d);$$

the divisor coordinates $\sigma^{(2)}$ are not separate elements of this ring. Remark 5.6 places its pushforward under the raw Novikov map over the image of that reduced ring [14, Remarks 2.3 and 5.6]. In the external direct sum of Section 5.8.2, by contrast, the variables $s_j \in H^*(Z)$ are independent coordinates on the common comparison target, and the center parameter is identified as $\varsigma_j = \varsigma_j^\circ + s_j$ [14, (5.47)–(5.48)]. The divisor components of these independent target coordinates retain the curve-class information lost by (2.4).

Lemma 2.3 (Faithful center base change). *Let $R_{Z,\text{src}}^{\text{red}}$ be the coefficient domain whose $\mathbf{C}[z]$ -extension is Iritani's divisor-equation-reduced source ring $\mathbf{C}[z][[Q_Z e^{\sigma^{(2)}}, \sigma']][\sigma^0]$. Let A_j be the numerical coefficient domain of the external direct sum in Section 5.8.2, with the independent target coordinates $t, s_0, \dots, s_{c-2}$ and with z omitted. The pullback determined by the center Novikov map and $\varsigma_j = \varsigma_j^\circ + s_j$ gives a continuous homomorphism*

$$\iota_j : R_{Z,\text{src}}^{\text{red}} \longrightarrow A_j.$$

On a combined divisor-equation generator it is

$$X_d = Q_Z^d \exp(\sigma^{(2)} \cdot d) \longmapsto Q^{i_* d} q_{\text{exc}}^{-\rho_Z \cdot d / (c-1)} \exp((\varsigma_j^\circ, (2) + s_j^{(2)}) \cdot d). \quad (2.5)$$

This map is injective. In particular, it induces a field embedding

$$\text{Frac}(R_{Z,\text{src}}^{\text{red}}) \hookrightarrow \text{Frac}(A_j).$$

After this field extension, each center summand in Iritani's comparison is the faithful scalar extension, followed by the formal coordinate change ς_j , of the intrinsic generic numerical even QDM of Z .

Proof. We prove injectivity by filtering to finitely many curve classes and using the independence of their exponentials in the target divisor coordinates. Choose integral divisors $D_1, \dots, D_\rho$ whose pairings separate $N_1(Z)$, and use their dual coordinates in $s_j^{(2)}$. The variables s_j occur only in the target A_j , not as coefficients in $R_{Z,\text{src}}^{\text{red}}$. This asymmetry is essential.

Filter the source by an ample degree pulled back from Y , together with the ordinary bulk order. Every fibre of the untagged monomial map in a bounded piece is finite: the restriction of an ample divisor from Y is ample on Z , and its bounded slice of $\overline{NE}(Z)$ contains only finitely many lattice points. This filtration is respected by the target map because $H \cdot i_* d = (i^* H) \cdot d$. The fixed shift ς_j° and the translations of the nondivisor and unit coordinates are invertible formal changes of target coordinates, so they do not affect injectivity.

Suppose a nonzero source element maps to zero and take its first nonzero filtered part. Group its finitely many terms by their common untagged target monomial and by their nondivisor bulk monomial. On one nonzero group the remaining relation has the form

$$\sum_{d \in F} a_d \exp\left(\sum_{k=1}^{\rho} (D_k \cdot d) s_{j,k}\right) = 0, \quad a_d \neq 0,$$

over a characteristic-zero coefficient domain that does not contain the variables $s_{j,k}$. The exponent vectors are distinct. Choose an integral one-parameter restriction on which they remain distinct. Derivatives of orders $0, \dots, |F| - 1$ at the origin give a Vandermonde matrix, hence every a_d is zero, a contradiction. Thus ι_j is injective on every bounded quotient and, by separated completeness, on the full reduced source.

Remark 2.3 of [14] gives the intrinsic reduced source, and Remark 5.6 gives its image under the noninjective map (5.15). Equations (5.47)–(5.48) then pull that image into the external direct sum over A_j , where the variables s_j are independent, and identify the direct sum with the blowup module. Associativity of scalar extension therefore identifies the center summand with scalar extension directly along the injective composite ι_j . Passing to fraction fields is faithfully flat. Finally, the inverse of the combined formal coordinate change $(\tau, s_0, \dots, s_{c-2})$, supplied by Theorem 5.18(7) and Section 5.8.2, transports the same statement to the original blowup coordinates. $\square$

Proposition 2.4 (Marker formulas for the QDM comparisons). *Let an additive marker be preserved by the regular scalar extensions and formal coordinate changes described above, and by every recorded grading suspension if its block type retains grading. The projective-bundle and blowup QDM comparisons on even cohomology then supply (2.1) and (2.2); the contribution of the j -th center summand is evaluated using its separate data ω_j .*

Proof. Projective bundles. For a rank- r bundle, Iritani–Koto’s Theorem 5.1 gives r copies of the base QDM after Laurent extension [15, Theorem 5.1]. Its invertible joint bulk Jacobian gives independent unit coordinates on the r target factors. In the j -th factor, changing its unit coordinate adds an independent scalar $u_j I$ to Euler multiplication. For two finite spectra, the resultant of their translated characteristic polynomials is a nonzero polynomial in $u_i - u_j$, with leading coefficient one. Hence the spectra of distinct factors are disjoint at the generic point: their primary blocks do not fuse when the direct sum is formed. Regularity, restriction to even cohomology, the fixed splitting field, and suspension invariance therefore carry the marker blockwise across the comparison, and additivity gives (2.1). Tensoring the bundle by a line bundle, when needed for the global-generation convention, changes neither the projective bundle nor the intrinsic Novikov embedding [15, Remark 1.2 and Remark 5.2].

Blowups. For a codimension- c blowup, Iritani’s Theorem 5.18 gives one ambient QDM and $c - 1$ center QDMs [14, Theorem 5.18]. Parts (4), (6), and (7) give the comparison asymptotics, the leading reconstruction parameters, and the invertible combined bulk Jacobian, respectively. The independent unit coordinates supplied by part (7) give the same nonzero resultant argument, so the spectra of the ambient and individual center summands are pairwise disjoint generically. Regularity, restriction to even cohomology, the fixed splitting field, suspension invariance, and Lemma 2.3 for each center copy then give (2.2). Each comparison is evaluated over its own common coefficient ring; the proof never composes Laurent comparisons recursively. $\square$

The QDM comparisons need not identify the underlying elements of the block monoid. They identify their images under the additive marker w , whose additivity is expressed by the square

$$\begin{array}{ccc} \mathcal{L}_{\mathcal{T}} \times \mathcal{L}_{\mathcal{T}} & \xrightarrow{+} & \mathcal{L}_{\mathcal{T}} \\ w \times w \downarrow & & \downarrow w \\ A \times A & \xrightarrow{+} & A. \end{array} \quad (2.6)$$

Proposition 2.4 supplies the resulting equalities in A . Its blowup statement also retains the data distinguishing the center summands, as required by Theorem 2.2.

2.3 The rank-two formal-exponent marker

We now specialize the general construction to $A = \mathbf{N}$. The block type $\mathcal{T}_{\text{exp}}$ retains the connection, pairing, and centered Euler endomorphism on H^{ev} . It retains no absolute grading, marked bulk section, or loop framing. Let E be a generic even block with centered connection

$$z \partial_z y = A(z)y, \quad A(z) = z^{-1}N + A_0 + zA_1 + z^2A_2 + \cdots. \quad (2.7)$$

Assume $\text{rk } E = 2$, $N \neq 0$, and $N^2 = 0$. Then $L = \text{im } N = \ker N$ is a line. The canonical elementary modification along this line is

$$E^{\sharp} = \{s \in E : s \bmod z \in L\}. \quad (2.8)$$

For this modification to remove the pole, the regular coefficient A_0 must preserve L . Pairing horizontality supplies exactly that condition.

Lemma 2.5 (The regular coefficient preserves the nilpotent line). *Horizontality of the Poincaré pairing implies $A_0 L \subset L$.*

Proof. Separated spectral factors are orthogonal for the horizontal Poincaré pairing: coefficientwise horizontality kills every off-diagonal block by an invertible Sylvester operator. Hence the restriction to E is nondegenerate. Write it as $P(z) = P_0 + zP_1 + \dots$. Horizontality gives

$$z\partial_z P(z) + A(z)^T P(z) + P(z)A(-z) = 0.$$

The z^{-1} coefficient gives $N^T P_0 = P_0 N$. Thus L is isotropic and, in dimension two, $L^\perp = L$. The constant coefficient is

$$A_0^T P_0 + P_0 A_0 + N^T P_1 - P_1 N = 0.$$

Evaluating this identity on (x, x) , where $x \in L = \ker N$, gives $(A_0 x, x) = 0$. Hence $A_0 x \in L^\perp = L$. $\square$

Choose a frame with $L = \langle e_1 \rangle$ and $Ne_2 = \nu e_1$, $\nu \neq 0$. The modified lattice has frame e_1, ze_2 , so

$$A^\sharp = S^{-1}AS - S^{-1}z\partial_z S, \quad S = \text{diag}(1, z). \quad (2.9)$$

The term $z^{-1}N$ becomes regular. The only possible new pole is the E_{21} -coefficient of A_0 , which vanishes by Lemma 2.5. Thus $A^\sharp = R + zR_1 + \dots$. Define

$$\delta^\sharp(E) := (\text{tr } R)^2 - 4 \det R. \quad (2.10)$$

Proposition 2.6 (Invariance of the modified residue). *Under formal change of even bulk coordinates, the modified connection is regular in every base direction and*

$$\partial R = [G_\partial, R] \quad (2.11)$$

for a regular matrix G_∂ . Consequently $\partial(\delta^\sharp) = 0$. Regular connection isomorphisms preserve the domain and value of $\delta^\sharp$.

Proof. There are two steps. Flatness first preserves the nonzero square-zero orbit of N . After the elementary modification, it removes the possible base pole and gives the Lax equation for the residue. Write a centered base equation as $\partial y = (z^{-1}C_\partial + C_{\partial,0} + O(z))y$. The z^{-2} coefficient of flatness gives $[N, C_\partial] = 0$, and the trace of the z^{-1} coefficient gives $\text{tr } C_\partial = 0$; here the base equation has been scalar-centered. The commutant of a nonzero rank-one nilpotent 2×2 matrix is the coefficient-field span of I and N , so $C_\partial = q_\partial N$. The same z^{-1} coefficient gives

$$\partial N = -q_\partial N + [N, q_\partial A_0 - C_{\partial,0}].$$

Writing $B = q_\partial A_0 - C_{\partial,0}$, one has $\partial(N^2) = -2q_\partial N^2 + [N^2, B]$; uniqueness for the formal coefficient recursion preserves $N^2 = 0$, while the nonzero leading value prevents N from becoming the zero formal section. Thus formal bulk transport preserves the nonzero square-zero orbit and its image line. After modification, the only possible base pole is $z^{-1}kE_{21}$. If $R = \begin{pmatrix} a & \nu \\ c & d \end{pmatrix}$, the z^{-1} flatness equation is $kE_{21} + [R, kE_{21}] = 0$. Its diagonal entries are νk and $-\nu k$, so $k = 0$. The constant term of the flatness equation is now (2.11); trace and determinant are constant under infinitesimal conjugation. A regular isomorphism carries N , L , and the modified lattice to their counterparts and conjugates the residue. $\square$

Proposition 2.7 (Exponent interpretation). *If ρ_1, ρ_2 are the eigenvalues of R , their classes modulo $\mathbf{Z}$ are the formal exponent classes of the original centered block and*

$$\delta^\sharp = (\rho_1 - \rho_2)^2.$$

Proof. The elementary modification uses integral powers of z , so it preserves formal exponent classes modulo $\mathbf{Z}$. The modified connection is regular singular with residue R ; the regular-singular classification identifies the exponent classes with the residue eigenvalues modulo $\mathbf{Z}$ [19, Chapter 3]. The displayed identity is the quadratic discriminant formula. $\square$

For such a block E , write ρ_1, ρ_2 for the eigenvalues of its modified residue in the fixed algebraic closure $\overline{K}$ of the generic coefficient field. Define the marker on each block by

$$w_{\text{exp}}(E) = \begin{cases} 1, & \text{rk } E = 2, \ N \neq 0, \ N^2 = 0, \ [\rho_1] \neq [\rho_2] \text{ in } \overline{K}/\mathbf{Z}, \\ 0, & \text{otherwise,} \end{cases} \quad (2.12)$$

and extend it additively to $w_{\text{exp}} : \mathcal{L}_{\mathcal{T}_{\text{exp}}} \rightarrow \mathbf{N}$. Put $I_{\text{exp}}(Y) = w_{\text{exp}}(\mathcal{B}_{\mathcal{T}_{\text{exp}}}(Y))$. Propositions 2.6 and 2.7 make this a well-defined additive marker for Proposition 2.4: conjugation preserves the two residue eigenvalues, while changes of regular-singular lattice and recorded suspensions shift exponents only by integers. The condition $\delta^\sharp \neq 0$ is insufficient: it says only $\rho_1 \neq \rho_2$ and therefore also marks resonant blocks.

2.4 The cubic block

Only a rank-two block can contribute to I_{exp} . For a cubic threefold, the calculation below isolates the unique such candidate and then computes the residue of its elementary modification.

Let $X \subset \mathbf{P}^4$ be a smooth cubic threefold, let $P = H|_X$, and put $q = Q^\ell$ for the line class and $r = (3q)^{1/2}$. Beauville's formulas give [3, main theorem and (2.1)–(2.3)]

$$P \star P = P^2 + 6q, \quad P \star P^2 = P^3 + 15qP, \quad P \star P^3 = 6qP^2 + 36q^2. \quad (2.13)$$

Thus, in the basis $(1, P, P^2, P^3)$,

$$K_X = 2 \begin{pmatrix} 0 & 6q & 0 & 36q^2 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad G_X = \frac{1}{2} \text{diag}(3, 1, -1, -3), \quad (2.14)$$

and $z^2 \partial_z S = (K_X + zG_X)S$. This agrees with [7, Section 3].

Proposition 2.8 (Small-even cubic block reduction). *The characteristic and minimal polynomials of K_X are both $T^2(T^2 - 108q)$. After adjoining r , its eigenvalues are $6r, -6r, 0, 0$, and the zero generalized eigenspace is one rank-two Jordan block. A normalized gauge regular in z gives*

$$\begin{aligned} z^2 \partial_z \tilde{S}_+ &= (6r + O(z^2)) \tilde{S}_+, \\ z^2 \partial_z \tilde{S}_- &= (-6r + O(z^2)) \tilde{S}_-, \\ z^2 \partial_z \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix} &= [J_0 + zD_0 + z^2E_0 + O(z^3)] \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix}, \end{aligned} \quad (2.15)$$

where

$$J_0 = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix}, \quad E_0 = \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix}. \quad (2.16)$$

Proof. The matrix

$$C = \begin{pmatrix} 6r^3 & -6r^3 & 0 & -7r^2 \\ 7r^2 & 7r^2 & -2r^2 & 0 \\ 3r & -3r & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, \quad \det C = -486r^5,$$

puts K_X into

$$C^{-1}K_X C = \begin{pmatrix} 6r & 0 & 0 & 0 \\ 0 & -6r & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Partition as $1 \mid 1 \mid 2$. At each positive order in z , the off-diagonal gauge coefficient is the unique solution of a Sylvester equation between blocks with disjoint scalar spectra. The gauge and inverse are regular in z ; explicitly, it has the normalized form

$$A(z) = I + \sum_{n \geq 1} A_n z^n, \quad A_n \text{ block-off-diagonal.} \quad (2.17)$$

Direct multiplication through order z^2 gives (2.16). $\square$

For the zero block, $N = J_0 = 2E_{12}$. With $S = \text{diag}(1, z)$,

$$R_X = 2E_{12} + D_0 - \frac{8}{81}E_{21} - \text{diag}(0, 1) = \begin{pmatrix} -19/18 & 2 \\ -8/81 & 1/18 \end{pmatrix}. \quad (2.18)$$

Its characteristic polynomial is $(T + 1/6)(T + 5/6)$. Thus the two exponent classes differ by $2/3 \notin \mathbf{Z}$, so the block is marked. Equivalently,

$$\boxed{\delta^\#(X) = \frac{4}{9}}, \quad I_{\text{exp}}(X) = 1. \quad (2.19)$$

2.5 Vanishing on centers through dimension two

Proposition 2.9 (Vanishing on low-dimensional centers). *For every smooth projective Z of dimension at most two and every center summand specified by comparison data ω , $I_{\text{exp}}(Z; \omega) = 0$.*

Proof. Points and curves. A point has one rank-one block. For $\mathbf{P}^1$, the relation $p \star p = Qe^{t_p}$ gives two rank-one generic blocks.

Let C have genus $g \geq 1$. There is no nonconstant genus-zero map to C , so its even big quantum product is classical. With $\int_C p = 1$ and $a = 2 - 2g$, its centered even connection is

$$z\partial_z Y = \left[z^{-1}aE_{21} + \text{diag}(1/2, -1/2) \right] Y.$$

For $g = 1$, $N = 0$. For $g > 1$, the modification $S = \text{diag}(z, 1)$ gives residue $aE_{21} - \frac{1}{2}I$, of discriminant zero. Its two eigenvalues, and therefore its two exponent classes, are equal. Thus every curve has zero marker.

Surfaces with nef canonical class. Let S be a surface with K_S nef. For a quantum-product term with input degree b , non-scalar Euler degree $a \geq 2$, curve class d , and nonunit even bulk degrees $e_\nu \geq 2$, the dimension axiom gives output degree

$$a + b - 2c_1(S) \cdot d + \sum_\nu (e_\nu - 2) \geq b + 2.$$

The centered Euler operator strictly raises ordinary degree. Its only block is the whole even cohomology, of rank $b_0 + b_2 + b_4 \geq 3$, so its marker is zero.

Other surfaces. A minimal surface with non-nef canonical class is $\mathbf{P}^2$ or a ruled surface $\mathbf{P}_C(V)$ [4, Chapter VI]. The relation $H^{*3} = Q$ makes $\mathbf{P}^2$ generically semisimple. Formula (2.1) and the curve case give zero for ruled surfaces. Every nonminimal surface is obtained from a minimal one by point blowups; formula (2.2) adds only zero point terms. This proves intrinsic vanishing for every surface. Lemma 2.3 identifies every separate center summand with a faithful scalar extension of the corresponding intrinsic generic block, and the marker is unchanged by that extension. $\square$

2.6 The projective endpoint and the contradiction

Formula (2.1) applied to $X \times \mathbf{P}^1 = \mathbf{P}_X(\mathcal{O}_X^{\oplus 2})$ gives

$$I_{\exp}(X \times \mathbf{P}^1) = 2. \quad (2.20)$$

On the other hand, over the generic numerical Novikov field $\Lambda = \text{Frac } \mathbf{C}[[Q]]$, $QH^\bullet(\mathbf{P}^4) = \Lambda[H]/(H^5 - Q)$. At $Q \neq 0$, multiplication by $5H$ has five distinct eigenvalues after algebraic closure. Thus

$$I_{\exp}(\mathbf{P}^4) = 0. \quad (2.21)$$

Proof of Theorem 1.1. Proposition 2.9 makes every actual center correction zero in a weak factorization of smooth projective fourfolds. Hence Theorem 2.2 makes $I_{\exp}$ birationally invariant. If $X \times \mathbf{P}^1$ were rational, then

$$2 = I_{\exp}(X \times \mathbf{P}^1) = I_{\exp}(\mathbf{P}^4) = 0,$$

a contradiction. $\square$

3 Consequences

No new quantum calculation is needed in this section. We first apply the marker directly in dimension three, then transfer the stabilization theorem through Kuznetsov's flop, and finally combine it with known criteria for universal CH_0 -triviality.

Proof of Corollary 1.2. In dimension three, every center in projective weak factorization has dimension at most one. The curve part of Proposition 2.9 therefore makes $I_{\exp}$ birationally invariant. The generic quantum product of $\mathbf{P}^3$ is semisimple, so $I_{\exp}(\mathbf{P}^3) = 0$. A marked block gives $I_{\exp}(Y) > 0$, ruling out rationality. $\square$

Proof of Corollary 1.3. Kuznetsov's flop gives a birational map between rank-two projective bundles $\mathbf{P}_V(U)$ and $\mathbf{P}_X(E^*)$ over V and its associated Pfaffian cubic threefold X [18, Theorems 2.17–2.18]. Each projective bundle is birational to the product of its base with $\mathbf{P}^1$. Hence $V \times \mathbf{P}^1$ is birational to $X \times \mathbf{P}^1$, which is irrational by Theorem 1.1. $\square$

Proof of Corollary 1.4. Voisin produces a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds along which the primitive minimal class $\Theta^4/4!$ is algebraic, hence along which CH_0 is universally trivial [23, Theorem 4.5 and Lemma 4.6]. The projective-bundle formula passes universal CH_0 -triviality to the product, and Theorem 1.1 supplies its irrationality. $\square$

Proof of Corollary 1.5. The Fermat equation is a sum of five forms in separated single variables, so [9] gives universal CH_0 -triviality; the projective-bundle formula passes it to the product. Irrationality follows from Theorem 1.1. $\square$

Proof of Corollary 1.6. The cubics of [24, Theorem 3.3] carry unirational parametrizations of degrees two and three, and [24, Corollary 3.5] keeps both degrees on $X \times \mathbf{P}^1$. Parametrizations of coprime degrees give a decomposition of the diagonal and hence universal CH_0 -triviality [24, Section 1]. Theorem 1.1 gives irrationality. $\square$

The marker formula has the same additive shape as Bittner’s blowup relation [2, Theorem 3.1]. If a marker extends additively to the Bittner group, then

$$I(\mathrm{Bl}_Z Y) - I(\mathbf{P}_Z(N_{Z/Y})) = I(Y) - I(Z)$$

follows from $I(\mathrm{Bl}_Z Y) = I(Y) + (c - 1)I(Z)$ and $I(\mathbf{P}_Z(N_{Z/Y})) = cI(Z)$. The $\mathbf{N}$ -valued marker used here is not such a group-valued motivic measure: we have constructed neither subtraction nor multiplicativity. The analogy explains why the center bound is decisive, but it does not give a factorization through the Grothendieck ring.

All results above concern one product stabilization. No statement about a second or arbitrary stabilization is implicit.

AI assistance disclosure

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Numbered statements are cited from the exact versions listed here. Where both a journal and a preprint version appear, the numbering is that of the preprint.

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